ONLINE CLASSROOM CONTACT PROGRAMME

(Academic Session : 2025 - 2026)

Enthusiast COURSE

PHASE : ALL PHASE

TARGET : PRE-MEDICAL 2026

Test Type :**MAJOR** Test Pattern :**NEET (UG)**

TEST DATE : 22-05-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	2	2	1	2	2	1	4	2	3	1	1	2	4	3	2	2	3	4	1	3	4	2	2	2	1	4	1	1	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	2	4	1	2	1	2	1	2	1	2	4	2	4	2	3	3	2	2	2	1	4	1	2	1	3	4	3	2	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	2	3	3	1	1	1	2	3	2	2	3	1	3	1	1	2	1	4	4	4	2	2	4	3	1	1	2	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	2	4	1	2	3	3	2	3	1	2	1	4	4	3	1	3	1	2	3	4	1	4	4	4	4	2	2	2	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	3	3	1	3	2	3	4	3	1	2	2	2	3	1	2	4	1	3	2	1	4	2	4	4	1	4	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	2	4	3	4	4	2	2	2	1	4	3	4	3	3	2	3	2	3	1	4	1	3	4	1	2	3	1	1	2

HINT – SHEET

1. Ans (1)

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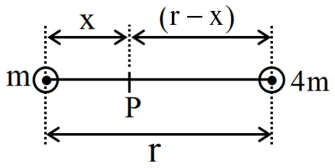
$(KE + PE)_{x=2} = (KE + PE)_{x=5}$

$0 + 6 = \frac{1}{2}mv^2 + 2$

$4 = \frac{1}{2}v^2$

$V = 2\sqrt{2}$ m/s.

2. Ans (2)



Say at point P (at distance 'x' from 'm'), the gravitational field is zero, then

$\frac{Gm}{x^2} = \frac{G(4m)}{(r-x)^2}$

$\Rightarrow x = \frac{r}{3}$

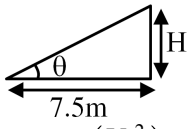
Now, gravitational potential at point P will be,

$V = \frac{-Gm}{x} - \frac{G(4m)}{(r-x)}$

$\Rightarrow V = \frac{-3Gm}{r} - \frac{12Gm}{2r}$

$= -\frac{9Gm}{r}$

3. **Ans (2)**



$\tan\theta = \left(\frac{V^2}{Rg}\right) = \frac{10^2}{150 \times 10} = \frac{1}{15}$

Also, $\tan\theta = \frac{H}{7.5}$

$\therefore \tan\theta = \frac{1}{15} = \frac{H}{7.5}$

$H = \frac{1}{2}$ m = 0.5 m

4. **Ans (1)**

Mass of removed disc = M

M.I of removed disc about O

$= \frac{1}{2}M\left(\frac{R}{3}\right)^2 + M\left(\frac{2R}{3}\right)^2 = \frac{1}{2}MR^2$ ... (i)

M.I of complete disc $= \frac{1}{2}(9M)R^2$ ... (ii)

$\therefore$ M.I of remaining disc

$= \frac{9}{2}MR^2 - \frac{1}{2}MR^2$

$= 4MR^2$

5. **Ans (2)**

$F \times \frac{3a}{4} = mg \times \frac{a}{2}$

$\Rightarrow F = \frac{2}{3}mg$

6. **Ans (2)**

Explain Question:

Find the maximum mass m to prevent system motion.

Concept:

Limiting Static Friction.

Solution:

Total mass on table (M) = 2 + 8 = 10kg

Maximum static friction (f_{max})= $\mu Mg = 0.5 \times 10 \times g = 5g$

Weight of hanging mass (W) = mg

For equilibrium: $W = f_{max}$

$mg = 5g \Rightarrow m = 5kg$

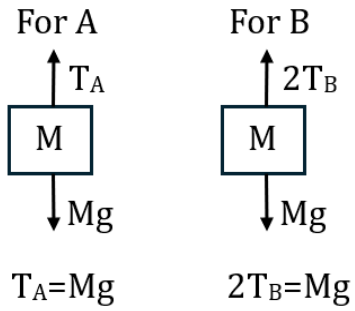
Final Answer: Option (2)

7. **Ans (1)**

Question Explanation: For given arrangement we need to find ratio of T_A and T_B

Concept: Translation equilibrium with pulley system

Solution:



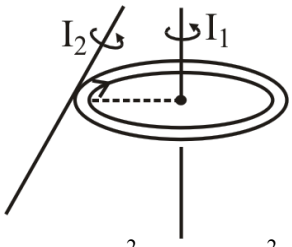
$T_B = \frac{Mg}{2}$

Now, $\frac{T_A}{T_B} = \frac{Mg}{Mg/2} = \frac{2}{1}$

$T_A : T_B = 2 : 1$

Final answer: option (1)

8. **Ans (4)**



$\because I_1 = MR^2 = 100$ kgm^2

So, $I_{cm} = \frac{MR^2}{2} = \frac{I_1}{2}$ [using perpendicular axis theorem]

using parallel axis theorem

$I_2 = I_{cm} + Md^2$

$= \frac{MR^2}{2} + MR^2 = \frac{3}{2}MR^2$

$= \frac{3}{2}(100)$

$= 150$ kgm^2

9. Ans (2)

1 atm = 10^5 pascal (approx)

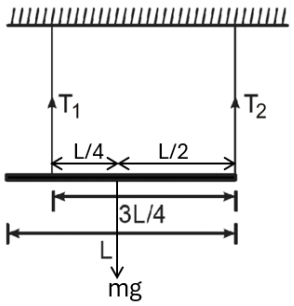
$$\frac{\Delta V}{V} = \frac{\Delta P}{B} = \frac{10 \times 10^5}{40 \times 10^9} = 2.5 \times 10^{-5}$$

10. Ans (3)

Explain question : Determine the ratio of tensions T_1 and T_2 in two strings supporting a uniform rod in horizontal equilibrium.

Concept : This question is based on the system is in equilibrium, and the sum of torques about any point should be zero, which leads to a relationship between T_1 and T_2.

Explanation :



From the torque balance equation

$\tau_C = 0$

$$T_1\left(\frac{L}{4}\right) = T_2\left(\frac{L}{2}\right)$$

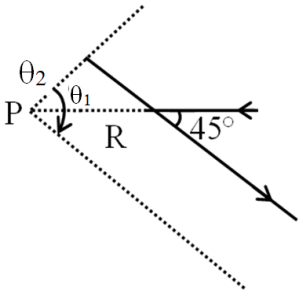
$$\frac{T_1}{T_2} = \frac{2}{1}$$

Final answer :- 3

11. Ans (1)

$\theta_1 = 90°$

$\theta_2 = 45°$



$$B = \frac{\mu_0 I\sqrt{2}}{4\pi R}(\sin 90° - \sin 45°)$$

$$B = \frac{\mu_0 I\sqrt{2}}{\mu\pi R}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

$$B = \frac{\mu_0 I}{4\pi R}(\sqrt{2}-1)$$

12. Ans (1)

Above curie temperature ferromagnetic substance converts into paramagnetic.

13. Ans (2)

Question Explanation: Calculating potential difference across a circuit segment with mixed components.

Concept: Kirchhoff's Voltage Law (KVL).

Solution:

To find $V_B - V_A$, we trace the potential from point A to point B.

Resistor (R) : Moving with the current, the potential drops by I ×R.

Battery (V): Moving from negative to positive terminal, the potential increases by 15 V .

Inductor (L) : The induced EMF is given by $V_L = L \times \frac{dI}{dt}$. Since the current is decreasing, $\frac{dI}{dt}$ is negative. Moving with the current through an inductor with decreasing current causes a potential increase (the inductor acts like a source to oppose the decrease).

Calculation:

Drop across Resistor (VR): $V_R = I \times R = 5A \times \Omega = 5V$ (drop)

Potential across Battery (V_{bat}): $V_{bat} = +15$ V (gain)

Induced EMF in Inductor (V_L):

$V_L = L\left|\frac{dI}{dt}\right| = (5 \times 10^{-3}H) \times (10^2 A/s) = 5V$ (gain due to decreasing current)

Tracing A to B:

$V_A - V_R + V_{bat} + V_L = V_B$

$V_B - V_A = -5 + 15 + 5 = 15V$

Final Answer: Option (2)

14. Ans (4)

When a conductor moves in a magnetic field in such a way that it cuts the magnetic lines of force, then a potential difference is developed between its two ends. In the given problem, this is only possible when the conductor moves along M.

15. Ans (3)

$R_{net} = 40 + 40 = 80\ \Omega$

$X_L = 100\ \Omega$

$Z = \sqrt{(80)^2 + (100 - 40)^2}$

$X_C = 40\ \Omega$ $\quad$ $Z = 100\ \Omega$

Power factor = $\cos\phi = \frac{R}{Z} = \frac{80}{100} = 0.8$

16. Ans (2)

Explain the Question : To find impedence and phase difference between 'V' and 'I' in RLC circuit.

Given :

$R = X_L = 2X_C$

Concept : Series RLC circuit

$Z = \sqrt{R^2 + (X_L - X_C)^2}$ and $\tan\phi = \frac{X_L - X_C}{R}$

Solution : As we know for a series LCR circuit impedance $(Z) = \sqrt{R^2 + (X_L - X_C)^2}$ and $\tan\phi = \frac{X_L - X_C}{R}$

Given, $X_L = R$, $X_C = \frac{R}{2}$

$Z = \sqrt{R^2 + \left(R - \frac{R}{2}\right)^2} = \frac{\sqrt{5}R}{2}$

$\tan\phi = \frac{R - R/2}{R} = \frac{1}{2} \Rightarrow \phi = \tan^{-1}\left(\frac{1}{2}\right)$

Final Answer: (2)

17. Ans (2)

$(U_{av})_B = (U_{av})_E = \frac{1}{4}\epsilon_0 E_0^2$

$\frac{1}{4}\epsilon_0(2)^2 = 8.85 \times 10^{-12}\ J/m^3$

18. Ans (3)

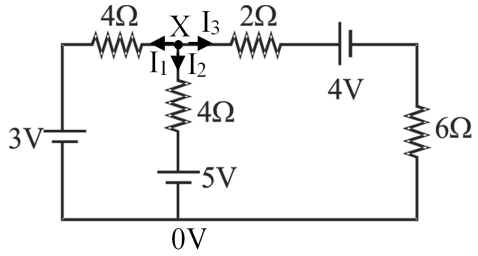
Use $R_g = \frac{I_s}{V_s}$

$Rg = \frac{10\text{ division}}{10^{-3}A} \times \frac{10^{-3}V}{2\text{ division}}$

$= \frac{10}{2}$

$= 5\Omega$

19. Ans (4)



Applying nodal method

$I_1 + I_2 + I_3 = 0$

$\frac{X-3}{4} + \frac{X-5}{4} + \frac{X-4}{8} = 0$

$\frac{2X - 6 + 2X - 10 + X - 4}{8} = 0$

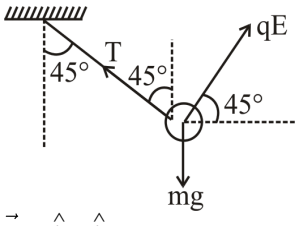
$5X = 20 \Rightarrow \boxed{X = 4V}$

$\Rightarrow I_3 = \frac{X-4}{8} = 0A$

20. Ans (1)

$F = q E_p = q\left[\frac{kp}{r^3}\sqrt{1 + 3\cos^2 45°}\right] = \sqrt{\frac{5}{2}}\frac{kqp}{r^3}$

21. Ans (3)



$\vec{E} = \hat{i} + \hat{j}$

$E = |\vec{E}| = \sqrt{2}N/c$

For equilibrium

$T\cos 45 + qE\sin 45 = mg$

$T\sin 45 = q E \cos 45 \Rightarrow T = qE$

$qE\cos 45 + q E \sin 45 = mg$

$\frac{qE}{\sqrt{2}} + \frac{qE}{\sqrt{2}} = mg$

$\sqrt{2}qE = mg$

$q = \frac{20}{2} = 10C$

22. Ans (4)

$c = \frac{\epsilon_0 A}{d - t\left(1 - \frac{1}{k}\right)} = \frac{\epsilon_0 A}{d - \frac{3}{4}d\left(1 - \frac{1}{k}\right)}$

$c = \frac{\epsilon_0 A}{d}\left(\frac{4k}{k+3}\right)$

23. Ans (2)

ΔP in decibles $= 10\log_{10}\left(\frac{200}{20}\right)$

$= 10$ db

24. Ans (2)

Velocity gradient $\longrightarrow (LT^{-1})(L^{-1}) = T^{-1}$

frequency $\longrightarrow T^{-1}$

time $\longrightarrow T$

25. Ans (2)

$k_1 = 3k$

$k_2 = \frac{3k}{2}$

$k_{eq} = \frac{9k}{2}$

$T = 2\pi\sqrt{\frac{2m}{9k}}$

26. Ans (1)

$|\Delta\vec{P}| = 2mv \sin 45^\circ = \sqrt{2}\, mv$

27. Ans (4)

We know that, $T = 2\pi\sqrt{\frac{L}{g}}$

$\frac{\Delta T}{T} = \frac{\Delta L}{2L} = \frac{1}{2}\alpha\Delta\theta = \frac{2\times10^{-6}\times10}{2} = 10^{-5}$

$\therefore \frac{\Delta T}{T}\times100 = 1\times10^{-5}\times100 = 10^{-3}\,\%$

28. Ans (1)

$\mu C_p\Delta T = \mu C_v\Delta T + \Delta W$

$\therefore \Delta W = \mu(C_p - C_v)\,\Delta T$

Fraction of heat converted into work

$= \frac{\Delta W}{\Delta Q} = \frac{\mu(C_p - C_v)\Delta T}{\mu C_p\Delta T}$

$= 1 - \frac{1}{\gamma} = 1 - \frac{3}{5} = \frac{2}{5}$

29. Ans (1)

$D = 2\times1 + 5\times\frac{10-9}{100} = 2.05$ cm

30. Ans (3)

$h = \frac{2T\cos\theta}{r\rho g} = \frac{(2)(0.465)(\cos135^\circ)}{(10^{-3})(13.6\times10^3)(9.8)} = 5\times10^{-3}$ m

$= 5$ mm

31. Ans (3)

Explain the question : Drops fall at equal intervals from a height of 18 m, and when the first drop hits the ground, the fourth just starts falling. We need the positions of the second and third drops.

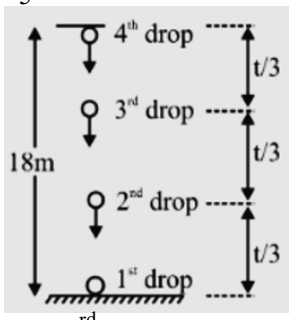
Concept : This question is based on motion under gravity, Free fall

time of fall $t = \sqrt{(2h/g)}$.

Solution :

$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times18}{10}} = \sqrt{\frac{18}{5}}$

$t = 3\sqrt{\frac{2}{5}}$



distance of 3rd drop from root : $\longrightarrow$

$s_3 = \frac{1}{2}\times10\times\frac{1}{9}\times9\times\frac{2}{5} = 2$m

distance of 2nd drop from root : $\longrightarrow$

$s_2 = \frac{1}{2}\times10\times\frac{4}{9}\times9\times\frac{2}{5} = 8$m

Final answer : 3

32. Ans (2)

$\frac{6}{Mo_2} = \frac{PV_0}{R(400)}$ …(1)

After leakage

$\frac{m}{Mo_2} = \frac{(P/2)V_0}{R(300)}$ …(2)

from (1) & (2)

$\frac{6}{m} = \frac{600}{400}$

$\Rightarrow m = 4g$

Gas leaked $= 6 - 4 = 2g$

33. Ans (4)

From given graph amplitude (a) = 1cm

Time period (T) = 8 sec

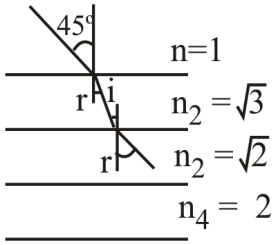
$\therefore \omega = \frac{2\pi}{8} = \frac{\pi}{4}$Hz

Acceleration $A = -\omega^2 a\sin\omega t$

at $t = \frac{4}{3}$sec, $A = -\frac{\pi^2}{16} \times 1 \times \sin\left(\frac{\pi}{4} \times \frac{4}{3}\right)$

$\Rightarrow A = \frac{-\pi^2}{16}\sin\left(\frac{\pi}{3}\right) \Rightarrow A = \frac{-\sqrt{3}}{32}\pi^2$ cm/s^2

34. Ans (1)



Applying snell's law in 1st & 3rd medium

$\sin 45^\circ = \sqrt{2}\sin r'$

$\frac{1}{\sqrt{2} \times \sqrt{2}} = \sin r'$

$r' = 30^\circ$

35. Ans (2)

Explain the Question : The wavelength of the photon emitted by a hydrogen atom is to be calculated when an electron makes a transition from n = 3 to n = 1. The required wavelength is obtained using the hydrogen spectral formula.

Concept : This question is based on hydrogen emission spectrum (Rydberg formula)

Formula :

$1/\lambda = (1/912\ Å)\ (1/n_1^2 - 1/n_2^2)$

Solution :

Here $n_1 = 1$ and $n_2 = 3$

$1/\lambda = (1/912)\ (1/1^2 - 1/3^2)$

$= (1/912)\ (1 - 1/9)$

$= (1/912)\ (8/9)$

$\lambda = (9 \times 912)/8$

$\lambda = 1026$ Å

Final Answer: (2)

36. Ans (1)

$m = \frac{f}{f-u} = \frac{-15}{-15+20} = \frac{-15}{5} = -3$

$V_{IM} = -m^2 V_{OM}$

$V_I = -(-3)^2\ (5) = -45$ cm/s

37. Ans (2)

$I = 4I_0 \cos^2\left(\frac{\phi}{2}\right)$

$\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}$

$I = 8\cos^2\left(\frac{\pi}{6}\right) = 6$ mW/m^2

38. Ans (1)

D = 80 , x = 20

$f = \frac{D^2 - x^2}{4D} = \frac{80^2 - 20^2}{4X80} = 18.75$ cm

39. Ans (2)

B.E of Helium = $(2m_P + 2m_N - m_{He})\ c^2$

= 28.4 MeV

40. Ans (1)

$n_2 = n_1\left[\frac{M_1}{M_2}\right]^a\left[\frac{L_1}{L_2}\right]^b\left[\frac{T_1}{T_2}\right]^c$ where $n_1 = a\ ms^{-2}$

$n_2 = a\left[\frac{1m}{2m}\right]^1\left[\frac{1s}{2s}\right]^{-2}$

$n_2 = a \times \frac{1}{2} \times 4 = 2a$

41. Ans (2)

According to their characteristics & worrking.

42. Ans (4)

For LED, in forward Bias, intensity increases with voltage.

43. Ans (2)

$_{69}A_3^{172} \xrightarrow{\alpha} {}_{71}A_2^{176}$

$_{71}A_2^{176} \xrightarrow{\alpha} {}_{73}A_1^{180}$

$_{73}A_1^{180} \xrightarrow{\beta} {}_{72}A^{180}$

44. Ans (4)

$$eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \quad ...(i)$$

$$\frac{eV}{4} = \frac{hc}{2\lambda} - \frac{hc}{\lambda_0} \quad ...(ii)$$

From equation (i) and (ii)

$$\Rightarrow 4 = \frac{\frac{1}{\lambda} - \frac{1}{\lambda_0}}{\frac{1}{2\lambda} - \frac{1}{\lambda_0}} \quad \text{On solving } \lambda_0 = 3\lambda$$

45. Ans (2)

$y = (A + B) \cdot \left(\overline{A \cdot B}\right)$

$= (A + B) \cdot \left(\bar{A} + \bar{B}\right)$

$= A\bar{A} + A\bar{B} + B\bar{A} + B \cdot \bar{B}$

$= 0 + A\bar{B} + B\bar{A} + 0$

= XOR Gate

46. Ans (3)

$CHCl_3$ + KOH cannot be not used to distinguish aniline and benzyl amine because the **carbylamine reaction** is given by both aniline (an aromatic primary amine) and benzyl amine (an aliphatic primary amine), making it impossible to use this reagent to distinguish between them.

47. Ans (3)

In option 3,

(A) $\longrightarrow$ **Chlorobenzene.**

(B) $\longrightarrow$ **No reaction.**

48. Ans (2)

$$pH = pK_a + \log \frac{[\text{Conjugate + base}]}{[\text{Acid}]}$$

$$= -\log (1.8 \times 10^{-5}) + \log \frac{[0.30]}{[0.20]}$$

pH = 4.92

pOH = 14 − pH

= 14 − 4.92 = 9.08

49. Ans (2)

Question Explanation:

The question asked for a mole fraction of benzene in vapour phase.

Concept:

Raoult's Law and Dalton's Low:

$$P_i = X_i \cdot P_i^0, \quad Y_i = \frac{P_i}{P_{total}}$$

Solution:

- Mole fraction in liquid:

$$X_{benzene} = \frac{1}{1+2} = \frac{1}{3}, \quad X_{toluene} = \frac{2}{3}$$

- Partial pressures:

$$P_{benzene} = \frac{1}{3} \cdot 42 = 14\text{mm}$$

$$P_{toluene} = \frac{2}{3} \cdot 36 = 24\text{mm}$$

- Total pressure = 14 + 24 = 38mm
- Mole fraction in vapour:

$$Y_{benzene} = \frac{14}{38} = \frac{7}{19}$$

Final Answer: Option(2)

50. Ans (2)

Question Explanation: For strong electrolytes, molar conductance (Λ_m) increases slightly with dilution due to decrease in inter-ionic attractions.

Concept: This question is based on Debye–Huckel–Onsager equation

Solution:

According to Debye–Hückel–Onsager equation:

$\Lambda_m = \Lambda_m^0 - A\sqrt{C}$

where Λ_m^0 = molar conductance at infinite dilution,

A = constant depending on nature of electrolyte and solvent,

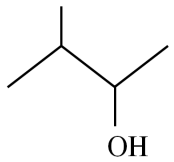
C = concentration.

The relation showing slow increase of Λ_m with dilution is $\Lambda_m = \Lambda_m^0 - A \cdot C^{1/2}$

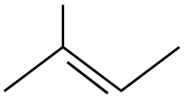
Final Answer: Option 2

51. Ans (1)

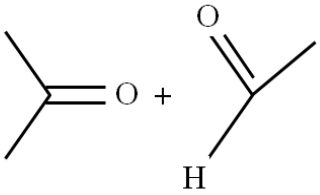
A =



B =



C =



52. Ans (4)

PCl_5 is reactive molecule because $P - Cl_{(axial)}$ is weaker than $P - Cl_{(equatorial)}$.

53. Ans (1)

(A) – (Q), (B) – (S), (C) – (P, Q), (D) – (Q)

B_2 & O_2 ⟶ Paramagnetic nature

Be_2 & H_2^{2-} ⟶ Doesn't exist

N_2^+ &N_2^- ⟶ Bond order = 2.5, Paramagnetic nature

O_2^+ &O_2^- ⟶ Paramagnetic nature

54. Ans (2)

A-II, B-I, C-IV, D-III

K ⟶ Violet

Ca ⟶ Brick Red

Sr ⟶ Crimson Red

Ba ⟶Apple Green

55. Ans (1)

Basic Radicals	Group Number
B. Cu^{+2}	II
A. Al^{+3}	III
D. Co^{+2}	IV
C. Ba^{+2}	V
E. Mg^{+2}	VI

Ans. (1) B, A, D, C, E

56. Ans (3)

Explain Question:

Identify the X^- in reaction.

Concept:

Redox reaction.

Solution:

$$2CuX_2 \xrightarrow[\text{temp}]{\text{room}} 2CuX + X_2$$

• The reaction shows spontaneous decomposition of Cu^{+2} to Cu^+ and anion to X_2.

• Cu^{+2} is reduced to Cu^+ hence X has to be strong reducing agent.

• CN^- and I^- are strong reducing agents:

$$2CuI_2 \longrightarrow 2CuI + I_2$$

$$2Cu(CN)_2 \longrightarrow 2CuCN + (CN)_2$$

• Reducing strength of halogens $F^- < Cl^- < Br^- < I^-$

• Hence the answer is (3) CN^- & I^-

Final Answer: Option (3)

57. Ans (4)

Lanthanum (L, Z = 57) actually has an electronic configuration of **$[Xe]5d^16s^2$** with zero electrons in its 4f orbital.

58. Ans (3)

Monodentate ligands do not show chelation.

59. Ans (2)

$Fe_4[Fe(CN)_6]_3$

Prussian blue only one type of ligand is present.

60. Ans (1)

Explain Question: Check assertion and reason about O and S

Concept: Melting, boiling point, Atomicity.

Solution: Oxygen exists as O_2 (diatomic) with weak intermolecular forces, while Sulfur exists mainly as S_8 (polyatomic), causing stronger intermolecular forces and higher m.p./b.p.; hence reason correctly explains assertion.

Final Answer : Option (1)

61. Ans (3)

According to Aufbau's principle, the 4f subshell starts filling only in the 6th period; the subshells filled in the 5th period are **5s, 4d, and 5p** only.

62. Ans (3)

By comparing the given equation with the standard Arrhenius form $\log_{10}k = \log_{10}A - \frac{E_a}{2.303RT}$, we get log10A = 5, which gives $A = 10^5$.

63. Ans (2)

Explanation the Question:

We need to calculate the de-Broglie wavelength of an electron in a hydrogen atom given its energy of separation and radius of the first orbit.

Concept:

This question is based on Bohr model & de-Broglie wavelength of electron.

Solution:

Given:

E = 3.4 eV (energy of electron), $r_1 = 0.53$ Å (radius of first orbit)

Formula:

$r_n = n^2 r_1$ (Bohr radius for nth orbit), $\lambda = \frac{2\pi r_n}{n}$ (de-Broglie wavelength)

Energy of electron in hydrogen:

$E_n = -\frac{13.6}{n^2}$eV (energy levels)

Given E = 3.4 eV, the excited state corresponds to:

$E_n = -3.4 \text{ eV} \Rightarrow n^2 = \frac{13.6}{3.4} = 4 \Rightarrow n = 2$

Radius of orbit:

$r_n = n_2 r_1 = 2^2 \cdot 0.53 = 2.12$Å

de-Broglie wavelength:

$\lambda = \frac{2\pi r_n}{n} = \frac{2\pi \cdot 2.12}{2} \approx 6.66$Å

Final Answer: (2)

64. Ans (3)

Since K_C is extremely large (3×10^6), the equilibrium strongly favors the product (NO_2), meaning the remaining concentration of either or both reactants must be much smaller.

65. Ans (3)

$xBrO_3^- + yCr^{+3} + zH_2O \rightarrow Br_2 + HCrO_4^- + H^+$

$3\overset{+5}{Br}O_3^- + 5Cr^{+3} \rightarrow \frac{3}{2}\overset{0}{Br_2} + 5H\overset{+6}{Cr}O_4^-$

$\left[3BrO_3^- + 5Cr^{+3} \rightarrow \frac{3}{2}Br_2 + 5HCrO_4^-\right] \times 2$

$22H_2O + 6BrO_3^- + 10Cr^{+3} \rightarrow 3Br_2 + 10HCrO_4^- + 34H^+$

$\downarrow \quad \downarrow \quad \downarrow$

z x y

x = 6, y = 10, z = 22

66. Ans (1)

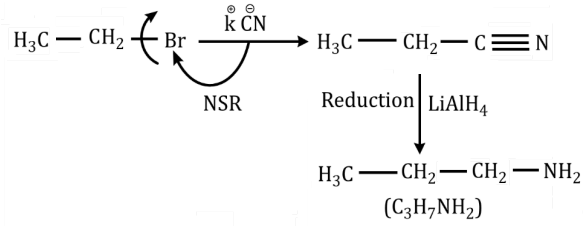
Since the transition from graphite to diamond is endothermic ($\Delta H > 0$) and its entropy decreases, graphite has lower energy and higher entropy, making it the more thermodynamically stable allotrope.

67. Ans (1)

Question Explanation: Identify x and y in the given sequence reaction.

Concept: NSR , Reduction

Solution:



So x ⇒ KCN

y ⇒ $LiAlH_4$

Final Answer - Option (1)

68. Ans (1)

Explaining the Question :- In acidic Medium, the electrophilicity of Carbonyl groups is ……

Concept:- This question is basedNAR of carbonyl compounds.

Solution :- In acidic medium, the electrophilicity of $>C=O$ group is <u>increased.</u> The carbonyl group can undergo protonation in acidic medium. When the oxygen atom of $>C=O$ group is protonated, if acquires a $\oplus$ve charge. The $\oplus$ve charge on oxygen make the $>C=O$ carbon even more electron deficient, there by increasing its electrophilicity and making it more susceptible to nucleophilic attack.

Final Answer :- (1)

69. Ans (2)

E2 will be favoured in option 1 whereas S_{N^2} will be favoured in option 2.

70. Ans (3)

Acidic solution turns blue litmus into red. For acidic solution millieq. of H^+ > millieq. of OH^-.

71. Ans (2)

$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$

$$\frac{14.35}{143.5} = 0.1 \text{ mol}$$

moles of NaCl = moles of AgCl = 0.1

moles of NaCl = $0.1 \times 58.5 = 5.85$ g

% of NaCl in the sample = $\frac{5.85}{11.7} \times 100 = 50\%$

72. Ans (2)

The compound is XeO_2F_2 (where $\frac{n_1}{n_2} = \frac{2}{2} = 1$), which has an sp^3d hybridized see-saw (folded square) geometry that is **non-planar** and polar with one lone pair.

73. Ans (3)

Phosphonic acid (H_3PO_3) contains only one P=O double bond, which means it has only **one $p\pi - d\pi$** bond rather than two.

74. Ans (1)

Both the assertion and reason are **true,** and the poor shielding effect of d-electrons leading to an increased effective nuclear charge correctly explains the general increase in second ionization enthalpy across the 3d series.

75. Ans (3)

Both Mn^{+3} and Co^{+3} act as strong **oxidizing agents**, as Mn^{+3} easily reduces to the stable d^5 configuration and Co^{+3} reduces to the stable t_{2g}^6 configuration.

76. Ans (1)

Explain Question: Predict the product of given reaction.

Concept: Kolbe electrolysis/ Isomerisation of alkane

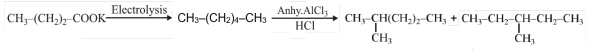
Solution:

Step 1: Kolbe Electrolysis.

Potassium butanoate ($CH_3(CH_2)_2COOK$) undergoes Kolbe Electrolysis to form a propyl radical, which dimerizes into n-hexane (X = $CH_3(CH_2)_4CH_3$).

Step 2: Isomerization.

Treating n-hexane with anhydrous $AlCl_3$/HCl causes isomerization, converting the straight-chain alkane into branched isomers: 2-methylpentane and 3-methylpentane (A and B).



Final Answer: Option (1)

77. Ans (1)

a ⟶ q, b ⟶ r, c ⟶ s, d ⟶ p


⟶ Homocylic aromatic


⟶ Anti aromatic


⟶ Non Aromatic


⟶ Heterocyclic aromatic

78. Ans (2)

$(CH_3)_3C^+$ has 9 α H

79. Ans (1)

Question Explanation:

Identify the purification/separation technique used to separate the various hydrocarbon fractions contained within crude oil.

Concept:

Separation Techniques, Distillation, Boiling Point Difference.

Solution:

Crude oil is a miscible mixture of hydrocarbons with close boiling points (B.P.).

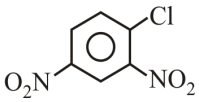
Separation of such a mixture requires high efficiency.

- Fractional distillation is the Correct method, utilizing a fractionating column to provide repeated vaporization and condensation cycles (theoretical plates).
- Simple distillation is only effective for mixtures with a large B.P. differences (=25°C or more).

The technique separates the various fractions (e.g., gasoline, kerosene) based on their B.P.

Final Answer: Option (1).

80. Ans (4)


IUPAC name : **1-chloro-2,4-dinitrobenzene**.

81. Ans (4)

In Lassaigne's test, when both nitrogen and sulfur are present together, they react with sodium to form NaSCN, which then combines with Fe^{3+} ions to form a blood-red complex.

82. Ans (4)

i-b, ii-c, iii-d, iv-a

dil. NaOH	⟶	Aldol reaction
Conc. NaOH	⟶	Cannizzaro reaction
$CHCl_3$ + aq. NaOH	⟶	Reimer-Tiemann reaction
SbF_3	⟶	Swarts reaction

83. Ans (2)

Number of subshells in a shell ⟶ n

Maximum number of electrons in a subshell ⟶2 $(2\ell + 1)$

Number of Radial Nodes ⟶ $n - \ell - 1$

Number of spectral lines for transition from energy level "n" to ground state ⟶ $\frac{n(n-1)}{2}$

84. Ans (2)

Statement C is incorrect because the decomposition of gaseous ammonia on a hot platinum surface at high pressure is a **zero-order reaction**, as the catalyst surface becomes completely saturated with gas molecules.

85. Ans (4)

$\because \Delta T_f = i \times K_f \times m$

$\Rightarrow 1.62 = \frac{i \times 4.9 \times \frac{2}{25} \times 1000}{122} \Rightarrow i = 0.504$

$\because \quad \alpha = \frac{1-i}{1-\frac{1}{n}} = \frac{1-0.504}{1-\frac{1}{2}}$

$\left[\because 2C_6H_5COOH \rightleftharpoons (C_6H_5COOH)_2\right]$

$= 0.496 \times 2$

$= 0.992 = 99.2\%$

86. Ans (3)

4

$[CrCl_2(OX)_2]^{-3}$

$[Co(en)_3]Cl_3$

$[Ir(acac)_2(H_2O)(NH_3)]^+$

$[Pt(Br)_2(NO_2)_2\ en]$

87. Ans (1)

 is the brown chromium complex intermediate formed during the **Etard reaction** when toluene is oxidized using chromyl chloride (CrO_2Cl_2) in CS_2.

88. Ans (1)

Glycine is a non-essential amino acid because the human body can synthesize it naturally, whereas valine, leucine, and histidine are essential amino acids that must be obtained through our diet.

89. Ans (2)

A negative standard electrode potential indicates that hydrogen gas is more stable than the species' reduced form, while a positive potential (greater than zero) means the reduced form is more stable than hydrogen gas.

90. Ans (2)

Looking at the pure components on the graph, the vapor pressure of pure A ($P°_A$) is higher than that of pure B ($P°_B$), making liquid A more volatile.

91. Ans (2)

Question Explanation: Identify whether Statement-I and Statement-II are correct.

Concept: Brain

Solution: The correct answer is option 2.

- Statement-I: Incorrect — Corpus callosum connects cerebral hemispheres, not cerebellar hemispheres (cerebellum has vermis connecting lobes).
- Statement-II: Incorrect — Brain stem consists of midbrain, pons and medulla oblongata, not entire hindbrain (hindbrain also includes cerebellum).

Final Answer: Option 2

92. Ans (2)

Question Explanation:

The question is based on resting membrane potential in neurons.

Concept:

Resting membrane potential – during resting stage, neuronal membrane shows selective permeability to ions.

Solution: At resting stage:

- Axonal membrane is more permeable to K^+ ions
- It is nearly impermeable to negatively charged proteins
- Na^+ permeability is comparatively low

Therefore, the correct statement is:

"Comparatively more permeable to K^+ ions and nearly impermeable to negatively charged protein."

Final Answer: Option (2)

93. Ans (4)

Question Explanation:

The question is based on phases of the menstrual cycle.

Concept: Menstrual cycle **Solution:**

- Menstrual flow occurs due to breakdown of endometrium ⟶ Correct
- Menstruation occurs if ovum is not fertilised ⟶ Correct
- Absence of menstruation may indicate pregnancy ⟶ Correct
- "Follicular phase is followed by menstrual phase" ⟶ Incorrect

Correct sequence:

- Menstrual phase ⟶ Follicular phase ⟶ Ovulation ⟶ Luteal phase

Final Answer: Option (4) The follicular phase is followed by the menstrual phase

94. Ans (1)

Question Explanation:

The question is based on hormonal control of parturition.

Concept:

Parturition – childbirth is regulated by neuroendocrine mechanisms and hormones.

Solution:

Statement-I: Correct

- Parturition is triggered by a complex neuroendocrine reflex involving hormones and uterine signals.

Statement-II: Correct

- Oxytocin stimulates uterine muscles and causes strong contractions during childbirth.

Final Answer: Option (1) Both Statement I and Statement II are correct

95. Ans (2)

Question Explanation: The question is based on different methods of contraception.

Concept: Barrier contraception – methods that physically prevent sperm from reaching the ovum.

Solution: Barrier methods from the given list are:

- Condoms
- Diaphragms
- Cervical caps
- Vaults

Not barrier methods:

- IUD ⟶ intrauterine device
- Pills ⟶ hormonal method
- Sterilisation ⟶ surgical method

Total barrier methods = 4 **Final Answer:** Option (2)

96. Ans (3)

Question Explanation: The question is based on different contraceptive methods and their functions.

Concept: Contraception

Solution:

- A. Condoms ⟶ Latex sheath ⟶ (IV)
- B. IUD's ⟶ Increase phagocytosis of sperms ⟶ (III)
- C. Sterilisation ⟶ Blocks gamete transport ⟶ (I)
- D. Saheli ⟶ Once-a-week pill ⟶ (II)

Thus:

- A–IV
- B–III
- C–I
- D–II

Final Answer: Option (3)

97. Ans (3)

Question Explanation:

The question is based on theories explaining the origin of life.

Concept:

Special creation theory states that Earth and all living organisms were created by a supernatural power.

Solution: According to the special creation theory:

- Earth is considered to be about 4000 years old
- All species were created in their present form

Other options represent scientific estimates of Earth's age.

Final Answer: Option (3) 4000 years

98. Ans (2)

Question Explanation: The question is based on methods used to determine the age of fossils.

Concept:

Radioactive dating – age determination using radioactive isotopes.

Solution

The age of fossils is commonly determined by Radio carbon dating (Carbon-14 dating).

- It measures decay of radioactive carbon isotopes present in fossil remains.
- Widely used for estimating ages of once-living materials.

Other options:

- Dendrochronology ⟶ age of trees
- Rock sediment counting ⟶ not accurate fossil dating
- Geological time scale ⟶ classification of eras, not direct dating method

Final Answer: Option (2) Radio carbon dating

99. Ans (3)

Question Explanation: The question is based on hormones of the adrenal gland and their functions during stress responses.

Concept: Fight or flight response – controlled by adrenal medullary hormones.

Solution: Hormones responsible for:

- Alertness
- Piloerection
- Increased blood glucose level

are catecholamines such as:

- Adrenaline (epinephrine)
- Noradrenaline (norepinephrine)

These are secreted by the adrenal medulla during stress or emergency situations.

Final Answer: Option (3) Catecholamines

100. Ans (1)

Question Explanation:

The question is based on the function of insulin in glucose regulation.

Concept:

Insulin – lowers blood glucose level by promoting uptake and utilization of glucose.

Solution:

Assertion: Correct

- Insulin is a hypoglycemic hormone because it decreases blood glucose level.

Reason: Correct

- Insulin increases cellular uptake and utilization of glucose.

Relation: Correct

- Increased glucose uptake by cells is the reason insulin lowers blood sugar.

Final Answer: Option (1) Both A and R are correct and R is the correct explanation of A

101. Ans (2)

Question Explanation:

The question is based on functions and disorders associated with thyroid hormones.

Concept:

Thyroxine regulates metabolism, growth, and development.

Solution:

- (A) Correct ⟶ Hypothyroidism during pregnancy affects normal development of the baby.
- (B) Correct ⟶ Thyroxine acts through intracellular receptors and does not require a secondary messenger.
- (C) Correct ⟶ Thyroxine regulates Basal Metabolic Rate (BMR).
- (D) Incorrect ⟶ Hypersecretion causes hyperthyroidism (high metabolism, weight loss, exophthalmia), not immunodeficiency.

Total correct statements = 3

Final Answer: Option (2)

102. Ans (1)

Question Explanation: The question is based on excretory products in different animals.

Concept: Uricotelism – excretion of nitrogenous waste mainly as uric acid with minimum water loss.

Solution:

Animals that excrete uric acid as pellets or paste conserve maximum water. This adaptation is common in:

- Reptiles
- Birds

These animals are called uricotelic animals.

Other groups:

- Mammals ⟶ mostly ureotelic
- Bony fish ⟶ ammonotelic
- Aquatic amphibians ⟶ ammonotelic

Final Answer: Option (1) Reptiles and Birds

103. Ans (4)

Question Explanation:

The question is based on reabsorption in the nephron.

Concept:

Proximal convoluted tubule – major site of selective reabsorption in the nephron.

Solution: In the PCT:

- Essential nutrients ⟶ reabsorbed completely
- Electrolytes ⟶ reabsorbed
- Water ⟶ reabsorbed extensively

Urea is not actively and completely reabsorbed like the above substances.

Final Answer: Option (4) Urea

104. Ans (4)

Question Explanation: The question is based on components of the immune system and their functions.

Concept:

Immune system – includes humoral and cell-mediated immunity.

Solution:

- (a) Antibody ⟶ Army of proteins ⟶ (iv)
- (b) B-lymphocytes ⟶ Humoral immune response ⟶ (ii)
- (c) T-lymphocytes ⟶ Cell-mediated immunity ⟶ (i)
- (d) T-cells help B-cells to produce them ⟶ H_2L_2 antibodies ⟶ (iii)

Thus:

- (a)–(iv)
- (b)–(ii)
- (c)–(i)
- (d)–(iii)

Final Answer: Option (4)

105. Ans (3)

Question Explanation:

The question is based on graft rejection and organ transplantation.

Concept:

Cell-mediated immunity plays a major role in graft rejection.

Solution:

- (a) Correct ⟶ Cell-mediated immunity causes graft rejection.
- (b) Incorrect ⟶ Grafts are not accepted from any source due to immune rejection.
- (c) Correct ⟶ Organs cannot be taken from just anybody.
- (d) Correct ⟶ Tissue matching and blood group matching are essential before transplantation.

Correct statements: (a), (c) and (d) **Final Answer:** Option (3)

106. Ans (1)

Question Explanation

The question is based on symptoms and diagnosis of typhoid fever.

Concept:

Typhoid fever – bacterial disease caused by Salmonella typhi.

Solution:

Statement-I: Correct

- Typhoid can be confirmed using the Widal test.

Statement-II: Correct

- Severe typhoid may lead to intestinal perforation and death if untreated.

Final Answer: Option (1) Both Statement I and Statement II are correct

107. Ans (3)

Question Explanation:

The question is based on transmission of infectious diseases through intravenous drug use.

Concept:

AIDS and Hepatitis B can spread through contaminated needles and syringes.

Solution:

People who take drugs intravenously using shared or infected needles are at high risk of:

- HIV infection causing AIDS
- Hepatitis B viral infection

These diseases spread through infected blood.

Final Answer: Option (3) AIDS and Hepatitis B

108. Ans (1)

Question Explanation

The question is based on cardiac output and heart sounds.

Concept:

Cardiac cycle

Solution:

Statement-I: Correct

- Cardiac output depends on:
  - Heart rate
  - Stroke volume

The body can alter both according to need.

Statement-II: Incorrect

- The two heart sounds ("lub" and "dub") are easily heard using a stethoscope.

Final Answer: Option (1) Statement-I is correct but Statement-II is incorrect

109. Ans (2)

Question Explanation

The question is based on events occurring during ventricular systole.

Concept:

Ventricular systole – contraction phase of ventricles during the cardiac cycle.

Solution:

During ventricular systole:

- Ventricular pressure increases rapidly
- This causes closure of:
  - Tricuspid valve
  - Bicuspid (mitral) valve

This prevents backflow of blood into the atria.

Final Answer: Option (2) Closure of tricuspid and bicuspid valves

110. **Ans (3)**

Question Explanation

The question is based on reproductive health and diseases affecting society.

Concept:

Sexually transmitted infections are infections spread mainly through sexual contact.

Solution:

- Infertility ⟶ medical condition
- MTP ⟶ medical termination of pregnancy
- Use of contraceptives ⟶ preventive measure
- STIs ⟶ spread infections and seriously affect public health

Thus, STIs are considered a major threat to a healthy society.

Final Answer: Option (3) STIs

111. **Ans (4)**

Question Explanation:

The question is based on developmental events during human gestation.

Concept: Human embryonic development **Solution:**

- 1 month ⟶ Foetal heart is formed ⟶ (iii)
- 2 month ⟶ Limbs and digits appear ⟶ (iv)
- 3 month ⟶ Most major organ systems are formed ⟶ (ii)
- 5 month ⟶ First movement of foetus felt ⟶ (i)

Thus:

- A–(iii)
- B–(iv)
- C–(ii)
- D–(i)

Final Answer: Option (4)

112. **Ans (1)**

Question Explanation:

The question is based on locomotion and characteristics of Nereis.

Concept: Annelid locomotion **Solution:**

- Assertion (A): Correct
  In Nereis, circular and longitudinal muscles help in locomotion.
- Reason (R): Incorrect
  Nereis possesses parapodia for swimming, but it is dioecious, not monoecious.

Final Answer: Option (1) (A) is true but (R) is false

113. **Ans (4)**

Question Explanation : The question asks to find the incorrect phylum–character–animal match.

Concept : This question is based on Animal Kingdom.

Solution : The incorrect statement is option (4)

Balanoglossus is a hemichordate, not a chordate, though it has bilateral symmetry.

- (1) – Correct – Branchiostoma (Cephalochordata) uses flame cells/solenocytes for excretion.
- (2) – Correct – Doliolum (Urochordata) is exclusively marine
- (3) – Correct – Pleurobrachia (Ctenophora) shows bioluminescene.

Final Answer : Option (4)

114. Ans (4)

Question Explanation

The question is based on identifying animal characteristics and symmetry.

Concept:

Radial symmetry with triploblastic body organization and coelom formation.

Solution:

Check the given animals:

Periplaneta americana ⟶ Bilateral

- symmetry
- Fasciola hepatica ⟶ Acoelomate, bilateral
- Pila ⟶ Bilateral symmetry

Echinus ⟶ Triploblastic, coelomate,

- complete digestive system, radial symmetry

Thus, the correct organism is Echinus.

Final Answer: Option (4) Echinus

115. Ans (4)

Question Explanation: The question is based on the hearing organ (tympanum) in vertebrates.

Concept: Tympanum acts as the ear drum and receives sound vibrations.

Solution:

- Bufo ⟶ tympanum present
- Hyla ⟶ tympanum present

Chameleon ⟶ reptiles possess tympanic

- structures for hearing

Hence, tympanum represents the ear in all the given animals.

Final Answer: Option (4) All of the above

116. Ans (4)

Question **Explanation:**

The question is based on respiratory volumes and capacities.

Concept:

Vital capacity – maximum volume of air that can be inspired after a forced expiration.

Solution:

Vital capacity (VC) is the maximum amount of air a person can breathe in after forcefully expiring all the air from the lungs.

VC = IRV + TV + ERV

Where:

- IRV = Inspiratory Reserve Volume
- TV = Tidal Volume
- ERV = Expiratory Reserve Volume

Final Answer: Option (4) Vital capacity

117. Ans (2)

Question **Explanation**

The question is based on transport of oxygen by blood.

Concept: Oxygen transport in blood **Solution:**

Under normal physiological conditions:

Every 100 mL of oxygenated blood carries

- about 20 mL of O_2
- Out of this, tissues utilize about 5 mL of O_2

Thus, oxygen delivered to tissues per 100 mL blood is approximately 5 mL.

Final Answer: Option (2) 5 mL

118. Ans (2)

Question Explanation

The question is based on types of joints in the human skeletal system.

Concept:

Ball and socket joint – allows movement in all directions.

Solution:

The head of the humerus fits into the glenoid cavity of the scapula to form the shoulder joint.

This joint permits:

- Rotation
- Flexion
- Extension
- Abduction and adduction

Hence, it is a ball and socket joint. **Final Answer:** Option (2) Ball and socket joint

119. Ans (2)

Question Explanation:

The question is based on disorders of the skeletal and muscular system.

Concept: Musculoskeletal disorders **Solution:**

- Gout ⟶ accumulation of uric acid crystals
- Osteoporosis ⟶ caused by decrease in estrogen level, not increase
- Tetany ⟶ muscle spasm due to low Ca^{2+} in body fluids
- Arthritis ⟶ inflammation of joints

Final Answer: Option (2) Osteoporosis - Increase level of estrogen

120. Ans (4)

Question Explanation:

The question is based on types and locations of epithelial tissues.

Concept:

Simple epithelium consists of a single layer of cells and is mainly involved in absorption, secretion, and diffusion.

Solution: Simple epithelium is found in:

- Lining of fallopian tube
- Tubular parts of nephron
- Air sacs (alveoli) of lungs

But the buccal cavity is lined by stratified squamous epithelium, not simple epithelium.

Final Answer: Option (4) Lining of buccal cavity

121. Ans (3)

Question Explanation

The question is based on mouth parts of Cockroach and their functions.

Concept: Biting and chewing mouthparts **Solution:** In cockroach:

- Mandibles are hard jaws used for:
  - Cutting/incising food
  - Grinding food

If mandibles are removed, both functions will be affected.

Final Answer: Option (3) Both

122. Ans (1)

Question Explanation: he question is based on the male reproductive system of Frog.

Concept: Male reproductive system in frog

Solution:

Path of sperms in male frog:

Testes ⟶ Vasa efferentia ⟶ Bidder's canal ⟶ Urogenital duct ⟶ Cloaca

- Sperms are produced in testes
- Passed through vasa efferentia
- Enter Bidder's canal of kidney
- Then move into urogenital duct
- Finally open into cloaca

Final Answer: Option (1)

123. Ans (3)

Question Explanation: The question is based on the logistic population growth equation.

Concept: Logistic growth
The logistic growth equation is: $dN/dt=rN(K-N/K)$
Where:

- (N) = Population density
- (K) = Carrying capacity
- (r) = Intrinsic rate of natural increase

Solution:

In the equation, r represents the organism's inherent capacity to grow under ideal conditions.

Final Answer: Option (3) Intrinsic rate of natural increase

124. Ans (3)

Question Explanation

The question is based on ecological interactions between organisms.

Concept: Ecological interactions **Solution:**

- A. Cuscuta on hedge plant ⟶ Parasitism ⟶ (III)
- B. Fig tree and female wasp ⟶ Mutualism ⟶ (I)
- C. Sparrow eating seeds ⟶ Predation ⟶ (II)
- D. Abingdon tortoise affected by goats ⟶ Competition ⟶ (IV)

Thus:

- A–III
- B–I
- C–II
- D–IV

Final Answer: Option (3)

125. Ans (1)

Question Explanation: The question is based on ecological relationships in ecosystems.

Concept: Ecosystem – interaction of organisms with each other and with the environment through energy flow and nutrient cycling.

Solution:

Food chains, food webs, and nutrient cycles are important because they:

- Maintain transfer of energy between trophic levels
- Recycle nutrients within the ecosystem
- Help sustain ecological balance

Other options are incorrect because energy is always lost as heat, organisms interact with environment, and all ecosystems function through these relationships.

Final Answer: Option (1) They help maintain the flow of energy and nutrients within the system

126. Ans (3)

Question Explanation:

The question is based on the source of energy in ecosystems.

Concept: Energy flow in ecosystem **Solution:**

In almost all ecosystems, the ultimate source of energy is sunlight. Producers capture solar energy through photosynthesis and pass it through the food chain.

Exception:

- Deep-sea hydrothermal ecosystems depend on chemical energy (chemosynthesis), not sunlight.

Final Answer: Option (3) Sunlight

127. **Ans (2)**

Question Explanation:

The question is based on biodiversity differences between the Western and Eastern Ghats.

Concept: Biodiversity hotspot **Solution:**

The Western Ghats show much greater diversity and endemism of:

- Amphibians

This is due to:

- High rainfall
- Humid climate
- Dense forests and isolated habitats

Final Answer: Option (2) Amphibians

128. **Ans (3)**

Question Explanation

The question is based on comparison of biodiversity between India and Scandinavian countries.

Concept:

Biodiversity includes genetic, species, and ecological diversity.

Solution: India has:

- Diverse climates
- Different landforms
- Varied ecosystems such as forests, deserts, grasslands, mountains, and wetlands

Therefore, India possesses much greater ecological diversity compared to Scandinavian countries.

Final Answer: Option (3) India has greater ecological diversity

129. **Ans (4)**

Question Explanation: The question is based on the history of cell biology.

Concept: Cell theory states that all living organisms are composed of cells and cells are the basic structural and functional units of life.

Solution: Cell theory was proposed by:

- Matthias Schleiden
- Theodor Schwann

Other scientists:

- Robert Brown ⟶ discovered nucleus
- George Palade ⟶ cell organelle studies
- Flemming ⟶ mitosis discovery

Final Answer: Option (4) Schleiden and Schwann

130. **Ans (3)**

Question Explanation: The question is based on the composition of plant cell wall.

Concept: Plant cell wall is mainly composed of cellulose, hemicellulose, and pectin.

Solution: Plant cell wall contains:

- Cellulose
- Hemicellulose
- Pectins

Chitin is absent in plant cell wall. It is found in:

- Fungal cell wall
- Exoskeleton of arthropods

Final Answer: Option (3)

131. Ans (1)

Question Explanation: The question is based on cell organelles included in the endomembrane system.

Concept: Endomembrane system includes membrane-bound organelles that work together in synthesis, packaging, and transport of materials.

Solution: The endomembrane system includes:

- Endoplasmic reticulum (ER)
- Golgi apparatus
- Lysosomes
- Vacuoles

It does not include:

- Mitochondria
- Chloroplasts
- Peroxisomes

Final Answer: Option (1) Endoplasmic reticulum (ER)

132. Ans (2)

Question Explanation: The question is based on types of chromosomes based on centromere position.

Concept: Chromosome morphology

Solution:

- Metacentric ⟶ centromere exactly in the middle
- Sub-metacentric ⟶ centromere slightly away from the middle
- Acrocentric ⟶ centromere near one end
- Telocentric ⟶ centromere at terminal end

Therefore, a chromosome with centromere slightly away from the middle is called sub-metacentric chromosome.

Final Answer: Option (2) Sub-metacentric chromosome

133. Ans (2)

Question Explanation: The question is based on phases of the cell cycle.

Concept: Cell cycle **Solution:** DNA replication and chromosome duplication occur during the S-phase (Synthesis phase) of interphase.

- G_1 phase ⟶ cell growth
- S phase ⟶ DNA replication
- G_2 phase ⟶ preparation for mitosis
- Metaphase ⟶ chromosome alignment during mitosis

Final Answer: Option (2)

134. Ans (2)

Question Explanation: The question is based on cell cycle and mitosis.

Concept: Cell cycle and Mitosis

Solution:

- (A) Correct ⟶ Growth and reproduction are characteristics of living cells.
- (B) Incorrect ⟶ Centriole duplication occurs in animal cells, not plant cells.
- (C) Correct ⟶ Prophase begins with condensation of chromosomal material.
- (D) Incorrect ⟶ Centromere splitting occurs in anaphase, not metaphase.

Correct statements = 2 **Final Answer:** Option (2)

135. Ans (3)

Question Explanation

The question is based on the endomembrane system of eukaryotic cells.

Concept:

Endomembrane system includes organelles whose functions are coordinated.

Solution:

Assertion: Correct

- Both endoplasmic reticulum and vacuoles are parts of the endomembrane system.

Reason: Incorrect

- Functions of organelles in the endomembrane system are coordinated with each other.

Final Answer: Option (3) Assertion is true but the Reason is false

136. Ans (1)

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137. Ans (2)

Question Explanation: The question is based on energy flow and productivity in ecosystems.

Concept: Ecosystem energetics

Solution:

- Plants capture about 2–10% of PAR ⟶ Correct
- Second law of thermodynamics explains continuous energy requirement ⟶ Correct
- Standing crop measured as biomass or number per unit area ⟶ Correct
- "First law of thermodynamics is not applicable to ecosystems" ⟶ Incorrect

The first law of thermodynamics (energy cannot be created or destroyed) is applicable to ecosystems.

Final Answer: Option (2)

138. Ans (4)

Question Explanation:

The question is based on population attributes vs individual organism characteristics.

Concept:

Population ecology – characteristics of a population

Solution:

- Sex ratio ⟶ It refers to the proportion of males and females in a population. It is a population-level attribute, not applicable to a single individual
- Birth rate ⟶ It is defined as the number of births per unit population per unit time. It applies only to a population, not an individual organism
- Death rate ⟶ It is the number of deaths per unit population per unit time. It is also a population-level parameter, not an individual trait

Since all the given options describe population-level characteristics:

Final Answer: Option (4)

139. Ans (1)

Question Explanation:

The question is based on internal structure of plant stem and root (anatomy of flowering plants).

Concept:

Dicot stem vs Monocot stem vs Root anatomy

Solution:

Let's analyze the given anatomical features:

- (a) Collenchymatous hypodermis ⟶ Characteristic of dicot stem
- (b) Sclerenchymatous pericycle in patches above phloem (semilunar patches) ⟶ Found in dicot stem
- (c) Vascular bundles arranged in a ring ⟶ A clear feature of dicot stem
- (d) Vascular bundles are conjoint and open ⟶ Presence of cambium indicates dicot stem (for secondary growth)

Final Answer: Option (1) Dicotyledonous stem

140. Ans (3)

Question Explanation: The question is based on the anatomy of dicot roots (internal structure of root in flowering plants).

Concept: Dicot root – xylem arrangement.

Solution:

In a typical dicotyledonous root like that of sunflower:

- The xylem is arranged in a radial pattern at the center of the stele.
- The number of xylem bundles is usually limited (diarch, triarch, tetrarch, etc.) depending on the species.
- In sunflower root, the condition is typically tetrach (tetrarch) — meaning four xylem bundles are present.

Final Answer: Option (3) Four

141. Ans (2)

Question Explanation: The question is based on aerobic respiration – Krebs cycle (citric acid cycle).

Concept: Decarboxylation in Krebs cycle.

Solution:

In the complete oxidation of one molecule of acetyl CoA:

- The acetyl group (2C) enters the Krebs cycle.
- During one turn of the cycle, two decarboxylation reactions occur:
  - Isocitrate → α-ketoglutarate (CO_2 released)
  - α-ketoglutarate → succinyl-CoA (CO_2 released)

So, total CO_2 released via decarboxylation = 2 times per acetyl CoA.

Final Answer: Option (2) — 2

142. Ans (1)

Question Explanation: The question is based on respiratory quotient (RQ) in aerobic respiration.

Concept:

Respiratory Quotient (RQ)

RQ = CO_2 produced/O_2 consumed

For carbohydrates (e.g., glucose):

$C_6H_{12}O_6 + 6O_2$ $6CO_2 + 6H_2O$

So,

- CO_2 produced = 6
- O_2 consumed = 6
- RQ = 6/6 = 1

Solution:

Assertion: "Complete oxidation of carbohydrates in aerobic respiration RQ value is 1." True

Reason: "During complete oxidation of carbohydrates, equal amount of CO_2 is evolved and O_2 is consumed." True

Explanation Check

The reason correctly explains the assertion because equal CO_2 production and O_2 consumption directly leads to RQ = 1.

Final Answer: Option (1) Both Assertion & Reason are True and the Reason is the correct explanation of the Assertion

143. Ans (4)

Question Explanation: The question is based on enzymes and co-factors in protein digestion (carboxypeptidase).

Concept: Carboxypeptidase – zinc metalloenzyme.

Solution:

- Carboxypeptidase is a digestive enzyme secreted by the pancreas.
- It is a metalloenzyme that requires a zinc ion (Zn^{2+}) as a co-factor for its catalytic activity.
- The Zn^{2+} ion is essential for maintaining the active site structure and assisting peptide bond hydrolysis.
- If Zn^{2+} is removed, the enzyme loses its catalytic activity.

Final Answer: Option (4)

144. Ans (2)

Question Explanation: The question is based on the life cycle of homosporous pteridophytes.

Concept: Pteridophytes – Gametophyte stage (prothallus).

Solution:

In homosporous pteridophytes, spores germinate to form a gametophyte called
- prothallus.
- This prothallus is:
  - Small
  - Free-living (independent)
  - Multicellular
- It bears the sex organs:
  - Antheridia (male)
  - Archegonia (female)

Thus, both antheridia and archegonia are found on the independent multicellular prothallus.

Final Answer: Option (2) Independent multicellular prothallus

145. Ans (4)

Question Explanation: The question is based on types of sexual reproduction in algae.

Concept: Oogamy in algae.

Solution:

- Oogamy involves:
  - Large non-motile egg
    Small motile or non-motile male
  - gamete (depending on group)

Now analyze the given organisms:

Fucus ⟶ shows oogamy with non-motile egg and non-motile male gametes (non-
- flagellated sperms)
  Polysiphonia ⟶ shows oogamy with non-motile egg and non-motile male gametes
- (spermatia)
  Gelidium ⟶ generally shows oogamy with
- non-motile gametes

So, all three show oogamy involving non-motile gametes (especially male gametes are non-motile in red algae groups like Polysiphonia and Gelidium).

Final Answer: Option (4) 2 and 3 both

146. Ans (4)

Question Explanation: Based on taxonomic hierarchy and binomial nomenclature.

Concept: Taxa, rank and scientific naming.

Solution:

A ⟶ True (organisms occupy taxonomic
- position)
  B ⟶ True (same specific epithet can occur
- in different genera)
  C ⟶ False (Human = Homo sapiens, not
- "Homo species")
  D ⟶ False (animals, mammals, dogs are
- different taxonomic levels)

Final Answer: Option: (4) A and B

147. Ans (1)

Question Explanation: The question is based on disease–pathogen matching.

Concept: Microbial diseases and causative agents.

Solution:

- A. Citrus canker ⟶ Bacteria
- B. Malaria ⟶ Protozoa (Plasmodium)
- C. Rust of wheat ⟶ Fungi (Puccinia)
  D. Bovine spongiform encephalopathy ⟶
- Prions

Final Answer: Option (1) — A:S, B:P, C:Q, D:T

148. Ans (4)

Question Explanation: The question is based on exponential growth in population/biomass dynamics.

Concept: Exponential growth equation $w = w_0 e^{rt}$
Here, r represents the intrinsic rate parameter in exponential growth.

Solution:

- r ⟶ represents the relative growth rate
  It also indicates the ability of an organism to
- produce new biomass
  It reflects the efficiency index of growth
- under ideal conditions

So, all given descriptions correctly describe r. **Final Answer:** Option: (4) All of the above

149. Ans (4)

Question Explanation: The question is based on flower morphology – placentation and aestivation.

Concept: Floral parts arrangement.

Solution:

Statement I : "The arrangement of ovules within the ovary is known as aestivation." False

- Arrangement of ovules in ovary = Placentation
- Aestivation is arrangement of sepals/petals in floral bud

Statement II : "The mode of arrangement of sepals and petals in floral bud with respect to other members of the same whorl is known as placentation." False

- Arrangement of sepals/petals = Aestivation, not placentation
- Placentation refers to ovules in ovary

Final Answer : Option (4) Both Statement-I and Statement-II are False

150. Ans (3)

Question Explanation: The question is based on plant growth regulators and heterophylly.

Concept: Gibberellins and developmental heterophylly.

Solution:

Statement I : "Spraying sugarcane crop with IAA can increase yield by 20 tonnes per acre." Incorrect

- Sugarcane yield increase is due to gibberellins (GA_3), not IAA
- GA_3 promotes internode elongation, increasing stem biomass

Statement II : "Cotton and coriander represent developmental heterophylly." Correct

- **Developmental heterophylly = different leaf forms in juvenile vs adult stages**
- **Cotton and coriander show this phenomenon**

Final Answer : Option (3) Statement-I is incorrect but Statement-II is correct

151. Ans (3)

Question Explanation: The question is based on the historical development of photosynthesis concept.

Concept: Origin of photosynthesis studies.

Solution

- Julius von Sachs → Showed chlorophyll is located in chloroplasts and starch is formed during photosynthesis
- T.W. Engelmann → Action spectrum of photosynthesis using oxygen-releasing bacteria
- Jan Ingenhousz → Showed plants release oxygen in presence of sunlight
- Cornelius Van Niel → Proposed that photosynthesis is a light-dependent redox reaction where hydrogen from a donor reduces CO_2 to carbohydrates

Final Answer: Option (3) Cornelius Van Niel

152. Ans (2)

Question Explanation: Identify the incorrect statement.

Concept: Photosynthesis

Solution:

Tomato and bell pepper can be grown as greenhouse crops → Correct

C_3 plants have PEPcase enzyme → Incorrect (PEPcase is characteristic of C_4/CAM plants, not C_3)

Cyclic photophosphorylation helps meet ATP/NADPH balance → Correct

Energy used to pump protons to create gradient → Correct

Final Answer: Option (2)

153. Ans (4)

Question Explanation: The question is based on seed dormancy and ancient seed germination examples.

Concept: Seed viability under extreme conditions.

Solution:

Seeds of Lupinus arcticus were excavated
- from Arctic Tundra.

After a long dormancy period, they were
- successfully germinated and even flowered.

The estimated dormancy period recorded is
- about 10,000 years.

Final Answer: Option (4) 10,000 years

154. Ans (3)

Question Explanation: The question is based on structure of ovule and seed anatomy.

Concept: Parts of ovule and seed.

Solution:

Let's match:

A. Hilum ⟶ Junction between ovule (seed)
- and funicle ⟶ III

B. Perisperm ⟶ Residual persistent
- nucellus ⟶ I
- C. Chalaza ⟶ Basal part of ovule ⟶ II

D. Funicle ⟶ Stalk attaching ovule to
- placenta ⟶ IV

Final Answer: Option (3) — A-III; B-I; C-II; D-IV

155. Ans (4)

Question Explanation: The question is based on biomolecules and macromolecules in biology.

Concept: Biomacromolecules – carbohydrates, proteins, nucleic acids.

Solution:

Cellulose ⟶ Polysaccharide (carbohydrate)
- ⟶ biomacromolecule

Haemoglobin ⟶ Protein ⟶
- biomacromolecule

Phospholipid ⟶ Not considered a true biomacromolecule (it is a lipid, but not a
- polymer like proteins/carbohydrates)

So, both cellulose and haemoglobin are biomacromolecules.

Final Answer: Option: (4) Both 1 and 2

156. Ans (4)

Question Explanation: The question is based on light reaction of photosynthesis and location of water splitting complex.

Concept: Photosystem II and oxygen evolving complex (OEC).

Solution:

Water splitting (oxygen evolving complex)
- is associated with Photosystem II (PSII).
- PSII is located in the granal thylakoids.

The OEC is positioned on the inner (lumenal) side of the thylakoid membrane,
- where it releases protons into the lumen.

Final Answer: Option (4) On inner side of granal thylakoid membrane

157. Ans (2)

Question Explanation: The question is based on ATP yield from ETS during oxidation of acetyl CoA.

Concept: Electron Transport System (ETS) – oxidative phosphorylation.

Solution:

From 1 acetyl CoA (Krebs cycle products):

- 3 NADH ⟶ (3 * 3 = 9) ATP
- 1 $FADH_2$ ⟶ (2) ATP

Total ATP via ETS = 9 + 2 = 11 **Final Answer:** Option (2) — 11

158. Ans (2)

Question Explanation: The question is based on fusion of stamens (androecium condition) in flowering plants.

Concept: Types of staminal cohesion.

Solution:

Monadelphous condition ⟶ stamens united into one bundle ⟶ Chinarose (Hibiscus)

- ⟶ A–III

Diadelphous condition ⟶ stamens in two

- bundles ⟶ Pea ⟶ B–IV

Polyadelphous condition ⟶ stamens in

- many bundles ⟶ Citrus ⟶ C–I

Epiphyllous condition ⟶ stamens attached

- to perianth (sepals/petals) ⟶ Lily ⟶ D–II

Final Answer: Option (2) — A-III, B-IV, C-I, D-II

159. Ans (2)

Question Explanation: The question is based on phyllotaxy (arrangement of leaves on stem).

Concept: Types of phyllotaxy.

Solution:

- Alternate phyllotaxy ⟶ one leaf per node

Opposite phyllotaxy ⟶ two leaves arise at

- each node and lie opposite to each other

Whorled phyllotaxy ⟶ more than two

- leaves at a node

Given condition matches opposite phyllotaxy only.

Final Answer: Option (2) Opposite phyllotaxy

160. Ans (1)

Question Explanation: The question is based on base pairing rules in DNA structure.

Concept: Chargaff's rules of DNA composition.

Solution:

Erwin Chargaff proposed that in double-

- stranded DNA:
  - A = T
  - G = C

    Hence, purine = pyrimidine and ratios
  - are constant (=1 in paired bases)
- Other options:

  Wilkins ⟶ X-ray diffraction studies
  - of DNA

    Franklin ⟶ X-ray diffraction
  - photograph

    Griffith ⟶ Transformation
  - experiment in bacteria

Final Answer: Option (1) Erwin Chargaff

161. Ans (4)

Question Explanation: The question is based on the Meselson and Stahl experiment (semi-conservative DNA replication).

Concept: DNA replication in $^{14}N \longrightarrow {}^{15}N$ medium

Solution:

- E. coli doubling time ≈ 20 minutes
- Time given = 60 minutes ⟶ 3 generations

Now track DNA:

1. 0 min: 100% light (LL)
2. 20 min: 50% hybrid (HL), 50% hybrid
3. 40 min: 25% hybrid, 75% hybrid

   60 min: all DNA becomes hybrid or heavy,
4. no fully light DNA remains

So, pure light DNA disappears after first replication cycle and does not persist in later generations.

Final Answer: Option (4) 0%

162. Ans (3)

Question Explanation: The question is based on types and functions of RNA.

Concept: mRNA, tRNA, rRNA, snRNA

Solution:

mRNA is made up of 80 different types of protein → False (mRNA is made of nucleotides, not proteins; it codes for
- proteins)

Final Answer: Option (3)

163. Ans (4)

Question Explanation: The question is based on gene regulation in prokaryotes (lac operon concept).

Concept: Inducible operon (lac operon).

Solution:

(A) Expression of genes is regulated by metabolic/physiological/environmental
- conditions → True

(B) Galactose acts as inducer for lac operon
- → False (inducer is lactose/allolactose)

(C) Operator region is adjacent to promoter
- in operons → True

(D) Lactose enters cell via permease →
- True

(E) Repressor regulation is called positive regulation → False (it is negative
- regulation)

Final Answer: Option (4) Only (A), (C) and (D)

164. Ans (3)

Question Explanation: The question is based on genetic terminology related to alleles in diploid organisms.

Concept: Allele condition in chromosomes.

Solution:

Hemizygous → presence of only one allele of a gene in a diploid organism (e.g., X-
- linked genes in males)

Final Answer: Option (3) Hemizygous

165. Ans (3)

Question Explanation: The question is based on Mendelian inheritance and expression of recessive traits.

Concept: Gene expression and enzyme activity.

Solution:

Recessive traits are expressed when the
- dominant allele is absent.
- Usually, the recessive allele leads to:
  - Non-functional enzyme formation
  - OR no enzyme production at all

In both cases, the functional product is absent, so the recessive phenotype is expressed.

Final Answer: Option (3) Both (1) & (2)

166. Ans (2)

Question Explanation: The question is based on biochemical control of cholesterol synthesis (statins).

Concept: Statins and enzyme inhibition.

Solution:

Statement 1: Statins are blood cholesterol–
- lowering agents → True

Statement 2: They act by competitive inhibition of HMG-CoA reductase, the key
- enzyme for cholesterol synthesis → True

Final Answer: Option (2) Both statements are correct

167. Ans (3)

Question Explanation: The question is based on restriction enzymes and their recognition sequences.

Concept: Restriction endonuclease (Hind II).

Solution:

- Hind II is a restriction enzyme.

It recognizes a specific 6 base pair
- palindromic sequence in DNA.

It cuts the DNA at a specific site within this
- sequence.

Final Answer: Option (3)

168. Ans (2)

Question Explanation: The question is based on restriction enzymes used in recombinant DNA technology.

Concept: Restriction endonucleases vs non-enzymatic DNA elements.

Solution: Check each term:

- EcoRI ⟶ restriction enzyme
- HindIII ⟶ restriction enzyme
- BamHI ⟶ restriction enzyme
- EcoRV ⟶ restriction enzyme
- HindII ⟶ restriction enzyme
- Ori (origin of replication) ⟶ DNA sequence, not enzyme
- rop (regulatory of primer) ⟶ plasmid regulatory gene, not enzyme

So total restriction enzymes = 5 **Final Answer:** Option (2) Five

169. Ans (3)

Question Explanation: The question is based on techniques used in biotechnology for DNA analysis.

Concept: Separation of DNA fragments.

Solution:

- Polymerase Chain Reaction (PCR) ⟶ amplifies DNA, not for separation
- Recombinant DNA technology ⟶ gene manipulation, not separation
- Gel electrophoresis ⟶ used to separate DNA fragments based on size
- Western blotting ⟶ detects proteins, not DNA

Final Answer: Option (3) Gel electrophoresis

170. Ans (1)

Question Explanation: The question is based on mobile genetic elements (transposons) and their applications.

Concept: Transposons in genetic engineering and functional studies.

Solution:

- Transposons are DNA sequences that can move within the genome.
- They are widely used as mutagenic tools and for gene function analysis, including gene silencing studies
- They are not directly used in:
  - Autoradiography ⟶ detection technique
  - PCR ⟶ DNA amplification method
  - DNA sequencing ⟶ base determination technique

Final Answer: Option (1) Gene silencing

171. Ans (4)

Question Explanation: The question is based on gene therapy for ADA (adenosine deaminase) deficiency.

Concept: Ex vivo gene therapy and limitations of engineered lymphocytes.

Solution:

- In ADA deficiency gene therapy:
  - Patient's lymphocytes are taken and modified outside the body
  - A functional ADA gene is inserted using a retroviral vector
  - These modified cells are then infused back into the patient

However, the key reason for periodic infusion is:

- The genetically engineered lymphocytes are not immortal, so they die after some time and need replacement

Final Answer: Option (4) Genetically engineered lymphocytes are not immortal cells

172. Ans (1)

Question Explanation: The question is based on Human Genome Project (HGP) facts.

Concept: Genome sequencing timeline and features.

Solution:

- "The sequence of chromosome 1 was completed in 2003" ⟶ Incorrect
  - Human Genome Project draft was completed in 2003, but chromosome 1 was finalized later (2006)
- HGP was a mega project ⟶ True
- Bacteria and yeast were used as hosts (BAC/YAC systems) ⟶ True
- 22 autosomes + X and Y chromosomes were sequenced ⟶ True

Final Answer: Option (1) The sequence of chromosome 1 was completed in 2003

173. Ans (3)

Question Explanation: The question is based on DNA replication and transcription processes.

Concept: Replication fork and enzyme functions.

Solution:

Assertion (A): "DNA replication occurs within a small opening of the DNA helix, referred to as replication fork." Correct

- **Replication takes place at the replication fork, where DNA strands separate.**

Reason (R): "During transcription the discontinuously synthesised fragments of DNA are joined by DNA ligase." Incorrect

- Discontinuous fragments (Okazaki fragments) are formed in DNA replication, not transcription.
- DNA ligase joins Okazaki fragments during replication, not transcription.

Final Answer: Option (3) (A) is correct but (R) is not correct

174. Ans (4)

Question Explanation: The question is based on complementary base pairing in double-stranded DNA.

Concept: Chargaff's rule: A = T and G = C between complementary strands

Solution: Given strand:

- A = 15%
- T = 15%
- G = 40%
- C = 30%

In complementary strand:

- A pairs with T ⟶ A = 15%
- T pairs with A ⟶ T = 15%
- G pairs with C ⟶ G = 30%
- C pairs with G ⟶ C = 40%

Final Answer: Option (4) 15% A, 15% T, 30% G, 40% C

175. Ans (1)

Question Explanation: The question is based on biogas (gobar gas) production and microorganisms involved.

Concept: Anaerobic decomposition and methane formation.

Solution:

- Gobar gas plants work under anaerobic conditions.
- The main microorganisms responsible are methanogens (e.g., Methanobacterium).
- They convert organic waste into methane (CH_4), the main component of biogas.

Final Answer: Option (1) Methanogens

176. Ans (2)

Question Explanation: The question is based on origin of medicinal drugs used in biotechnology and pharmacology.

Concept: Cyclosporin A as an immunosuppressant.

Solution:

- Cyclosporin A is a powerful immunosuppressant drug used in organ transplantation to prevent rejection.
- It is produced by the fungus Trichoderma polysporum.

So, it is of fungal origin. **Final Answer:** Option (2) Fungus

177. Ans (3)

Question Explanation: The question is based on genetic inheritance patterns in F_2 generation.

Concept: Genotype vs phenotype ratios.

Solution:

- Complete dominance ⟶ Genotypic ratio ≠ Phenotypic ratio
- (Genotype 1:2:1, Phenotype 3:1)
- Incomplete dominance ⟶ Genotypic ratio = Phenotypic ratio
- (e.g., Snapdragon: 1:2:1)
- Co-dominance ⟶ Genotypic ratio = Phenotypic ratio
- (e.g., ABO blood group: 1:2:1 expression pattern)

Final Answer: Option (3) Both (1) & (2)

178. Ans (1)

Question Explanation: The question is based on Mendel's traits in garden pea and genotype–phenotype relationship.

Concept: Dominant vs recessive expression in heterozygous condition.

Solution:

- A heterozygous genotype always expresses the dominant trait, not the recessive one.
- Therefore, any option showing a recessive phenotype with "homozygous or heterozygous" is incorrect.

Now evaluate:

- (1) Flower position ⟶ Terminal
  - Terminal is recessive, so it cannot be expressed in heterozygous condition
  - Heterozygous would show axial ⟶ Incorrect match
- (2) Pod shape ⟶ Inflated (dominant, correct)
- (3) Pod colour ⟶ Green (dominant, correct)
- (4) Seed colour ⟶ Yellow (dominant, correct in Mendelian pea traits)

Final Answer: Option (1)

179. Ans (1)

Question Explanation: The question is based on Mendel's laws of inheritance (dominance and F_2 ratio).

Concept: Law of dominance and monohybrid cross outcomes.

Solution:

Statement I

"In a dissimilar pair of factors one member dominates the other." Correct

- In a heterozygote, dominant allele masks recessive allele.

Statement II

"Law of dominance explains the 3:1 ratio in F_2 generation." Correct

- The 3:1 phenotypic ratio arises due to expression of dominance in heterozygotes during segregation in F_2.
- Dominance explains why only one trait is expressed in heterozygous condition, leading to 3 dominant : 1 recessive phenotype.

Final Answer: Option (1) Both statement-I and statement-II are correct

180. Ans (2)

Question Explanation: The question is based on lac operon regulation in E. coli.

Concept: Catabolite repression (glucose effect on lac operon).

Solution:

- When glucose is abundant, E. coli prefers glucose as energy source.
- This leads to catabolite repression ⟶ lac operon remains inactive.
- As a result, structural genes (lac Z, Y, A) are not expressed.
- Lactose is not required, so operon is switched off.

Now evaluate options:

- Structural gene will be expressed ⟶ False
- Structural gene will not be expressed ⟶ True
- Regulatory gene will not be expressed ⟶ False (regulatory gene is constitutive)
- Repressor protein will not bind due to lactose ⟶ False (lactose absent)

Final Answer: Option (2) Structural gene will not be expressed